HW04 · Ch 24 Electric Potential — Worked Solutions

Source: CST homework04 (Chapter 24), mxhuang@scut.edu.cn · study-slides (vetame21.github.io/study-slides)

1 If 500 J of work are required to carry a 40 C charge from one point to another, the potential difference between these two points is:

A) 12.5 V ✓ (answer)
B) 20,000 V
C) 0.08 V
D) depends on the path
E) none of these

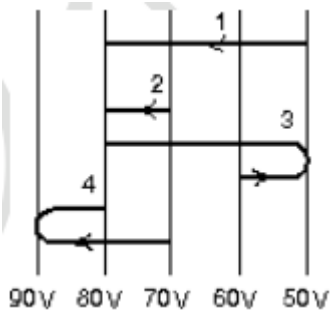
Answer: A

SOLUTION

$W = q\,\Delta V \Rightarrow \Delta V = \frac{500}{40} = 12.5$ V. The electric force is conservative, so the path doesn't matter.

---

2 An electron goes from one equipotential surface to another along one of the four paths shown. Rank the paths according to the work done by the electric field, from least to greatest.



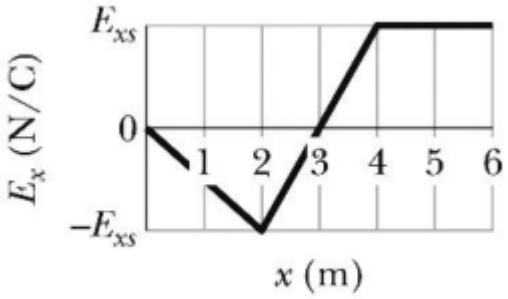
A) 1, 2, 3, 4
B) 4, 3, 2, 1
C) 1, then 3, then 4 and 2 tie
D) 4 and 2 tie, then 3, then 1 ✓ (answer)
E) 4, 3, 1, 2

Answer: D

SOLUTION

Work by the field depends only on the start and end potentials: $W = -q\,\Delta V = e(V_f - V_i)$ for an electron. Paths 2 and 4 start and end on the same pair of surfaces (a tie); ranking $e(V_f - V_i)$ gives 4 = 2, then 3, then 1.

3 **The graph shows E_x as a function of position. If $E_{xs} = 100$ N/C, what is the potential difference $V(6\,\text{m}) - V(3\,\text{m})$?**



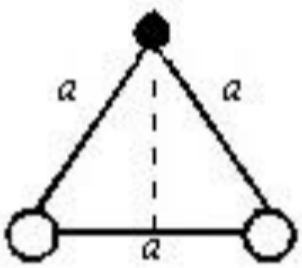
A) 250 V
B) 50 V
C) 0 V
D) −50 V
E) −250 V ✓ (answer)

Answer: E

SOLUTION

$\Delta V = -\int_3^6 E_x\,dx = -(\text{area under the graph from 3 to 6})$. The area is a triangle (3→4 m, rising 0→100) plus a rectangle (4→6 m at 100): $50 + 200 = 250$, so $\Delta V = -250$ V.

4 **Two particles with charges Q and $-Q$ are fixed at two vertices of an equilateral triangle with side a. The work required to move a charge q from the third vertex to the midpoint of the line joining the fixed charges is:**



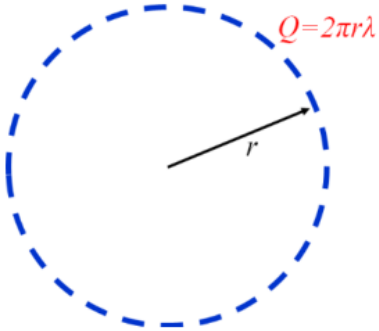
A) 0 ✓ (answer)
B) kQq/a
C) kQq/a^2
D) $2kQq/a$
E) $\sqrt{2}\,kQq/a$

Answer: A

SOLUTION

Both points are equidistant from $+Q$ and $-Q$, so $V = k\frac{Q}{r} - k\frac{Q}{r} = 0$ at each. Same potential, so $W = q\,\Delta V = 0$.

5 A wire with charge density λ C/m is bent into a circle of radius r. What is the electric potential at the center of the circle?



A) $\lambda/4\pi\varepsilon_0 r$
B) $\lambda/4\pi\varepsilon_0$
C) $\lambda/4\varepsilon_0$
D) $\lambda/2\varepsilon_0$ ✓ (answer)
E) λ/ε_0

Answer: D

SOLUTION

Every bit of the ring is the same distance r from the center, so $V = \frac{q}{4\pi\varepsilon_0 r}$ with $q = 2\pi r\lambda$: $V = \frac{2\pi r\lambda}{4\pi\varepsilon_0 r} = \frac{\lambda}{2\varepsilon_0}$.

---

6 The electric potential in a region is $V = -7.5x^2 + 3x$ (V, m). In this region the equipotential surfaces are:

A) planes parallel to the x axis
B) planes parallel to the yz plane ✓ (answer)
C) concentric spheres centered at the origin
D) concentric cylinders with the x axis as the cylinder axis
E) unknown unless the charge is given

Answer: B

SOLUTION

V depends only on x, so V is constant wherever x is constant: the planes $x = \text{const}$, which are parallel to the yz plane.

7 **The electric potential is $V = -7.5x^2 + 3x$ (V, m). What is the electric field?**

A) $(15x - 3)\,\hat{\imath}$ ✓ (answer)
B) $(-15x + 3)\,\hat{\imath}$
C) $(-2.5x^3 + 1.5x^2)\,\hat{\imath}$
D) $(2.5x^3 - 1.5x^2)\,\hat{\imath}$
E) 0

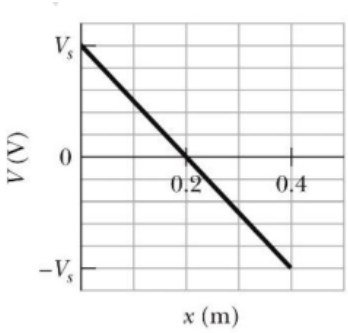
Answer: A

SOLUTION

$E_x = -\frac{dV}{dx} = -(-15x + 3) = 15x - 3$. Don't forget the minus sign: option B is the derivative without it.

---

8 **The graph shows $V(x)$ in a region. What is E_x if $V_s = 50$ V?**



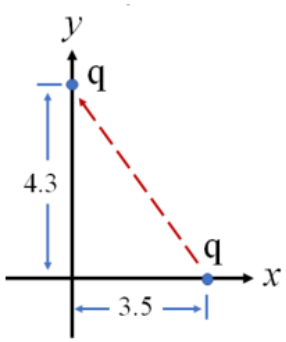
A) 250 V/m ✓ (answer)
B) 40 V/m
C) 10 V/m
D) –40 V/m
E) –250 V/m

Answer: A

SOLUTION

The graph is a straight line falling from 50 V at $x = 0$ to 0 at $x = 0.2$ m, a slope of -250 V/m. $E_x = -\frac{dV}{dx} = +250$ V/m.

9 **A charge 5.5×10^{-8} C is fixed at the origin. A charge -2.3×10^{-8} C is moved from $x = 3.5$ cm to $y = 4.3$ cm. The change in potential energy of the system is:**



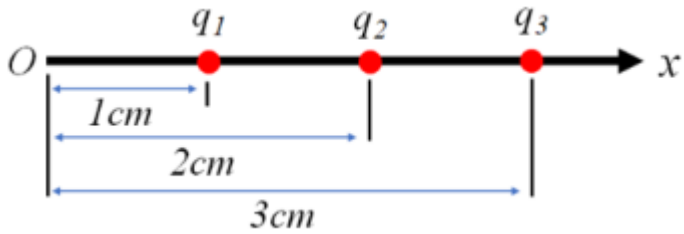
A) 3.1×10^{-5} J
B) -3.1×10^{-5} J
C) 6.0×10^{-5} J ✓ (answer)
D) -6.0×10^{-5} J
E) 0 J

Answer: C

SOLUTION

$\Delta U = q\,(V_f - V_i) = (-2.3 \times 10^{-8})(9 \times 10^{9})(5.5 \times 10^{-8})\left(\frac{1}{0.043} - \frac{1}{0.035}\right) = +6.0 \times 10^{-5}$ J.
Pulling the negative charge farther from the positive one raises U.

10 **Particles on the x axis: 1×10^{-8} C at 1 cm, 2×10^{-8} C at 2 cm, -3×10^{-8} C at 3 cm. The potential energy of the arrangement (zero at infinite separation) is:**



A) $+4.9 \times 10^{-4}$ J
B) -4.9×10^{-4} J ✓ (answer)
C) $+8.5 \times 10^{-4}$ J
D) -8.5×10^{-4} J
E) 0 J

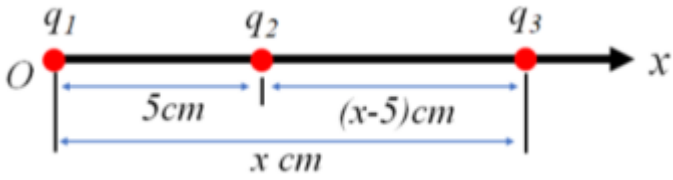
Answer: B

SOLUTION

$$U = k\left[\frac{(1)(2)}{0.01} + \frac{(1)(-3)}{0.02} + \frac{(2)(-3)}{0.01}\right] \times 10^{-16} = (9 \times 10^{9})(-5.5 \times 10^{-14}) = -4.9 \times 10^{-4}\ \text{J}.$$

Note that particles 1 and 3 are 0.02 m apart. **[NOTE]** The official solution writes 0.01 m for this pair, but its final answer, -4.9×10^{-4} J, matches the correct 0.02 m. **Why this method:** the energy of a system is the sum over every pair, kq_iq_j/r_{ij}.

11 **Two identical particles, each with charge q, are on the x axis at the origin and at $x = 5$ cm. A third particle, $-q$, is placed on the x axis so that the system's potential energy is zero. Its x coordinate is:**



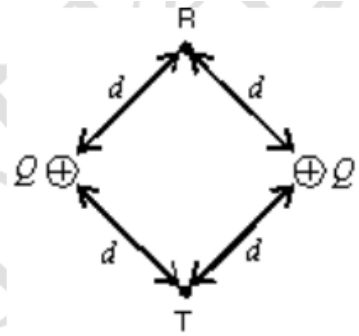
A) 13 cm ✓ (answer)
B) 2.5 cm
C) 7.5 cm
D) 10 cm
E) –5 cm

Answer: A

SOLUTION

$U = kq^2\left(\frac{1}{5} - \frac{1}{x} - \frac{1}{x-5}\right) = 0$ (cm) $\Rightarrow x^2 - 15x + 25 = 0 \Rightarrow x = \frac{15+\sqrt{125}}{2} = 13.1$ cm.

12 **Points R and T are each a distance d from each of two equal positive charges Q, as shown. The work required to move a charge q from R to T is:**



A) 0 ✓ (answer)
B) kQq/d^2
C) kQq/d
D) $kqQ/(\sqrt{2}d)$
E) $kQq/(2d)$

Answer: A

SOLUTION

At R and at T the potential is the same, $V = 2kQ/d$, so $W = q\,\Delta V = 0$.

13 **A 5 cm radius conducting sphere has surface charge density 2×10^{-6} C/m². Its potential (relative to far away) is:**

A) 1.1×10^{4} **V ✓ (answer)**
B) 2.2×10^{4} V
C) 2.3×10^{5} V
D) 3.6×10^{5} V
E) 7.2×10^{6} V

Answer: A

SOLUTION

$V = \frac{kq}{R}$ with $q = 4\pi R^2 \sigma$ gives $V = \frac{\sigma R}{\varepsilon_0} = \frac{(2\times10^{-6})(0.05)}{8.85\times10^{-12}} = 1.1 \times 10^{4}$ V.

---

14 **Two conducting spheres are far apart. The smaller carries total charge Q; the larger has twice the radius and is neutral. After they are connected by a wire, the charges on the smaller and larger spheres are:**

A) $Q/2$ and $Q/2$
B) $Q/3$ **and** $2Q/3$ **✓ (answer)**
C) $2Q/3$ and $Q/3$
D) 0 and Q
E) $2Q$ and $-Q$

Answer: B

SOLUTION

Connected conductors reach the SAME potential: $\frac{kq_1}{r} = \frac{kq_2}{2r} \Rightarrow q_2 = 2q_1$. With $q_1 + q_2 = Q$: $q_1 = Q/3$, $q_2 = 2Q/3$.

---

15 **A 5 cm radius isolated conducting sphere is charged to +100 V relative to far away. The charge density on its surface is:**

A) $+2.2 \times 10^{-7}$ C/m²
B) -2.2×10^{-7} C/m²
C) $+3.5 \times 10^{-7}$ C/m²
D) -3.5×10^{-7} C/m²
E) $+1.8 \times 10^{-8}$ **C/m² ✓ (answer)**

Answer: E

SOLUTION

Reverse of Q13: $\sigma = \frac{\varepsilon_0 V}{R} = \frac{(8.85\times10^{-12})(100)}{0.05} = 1.8 \times 10^{-8}$ C/m², positive because $V > 0$.

16 **For the potential $V = 3x^2z - 2yz^3$, what is the electric field at the point (2, 2, 2)?**

A) $-24\hat{\imath} + 16\hat{\jmath} + 36\hat{k}$ **✓ (answer)**

B) $12\hat{\imath} - 16\hat{\jmath} - 16\hat{k}$

C) $24\hat{\imath} - 16\hat{\jmath} - 36\hat{k}$

D) $-6\hat{\imath} + 2\hat{\jmath} + 6\hat{k}$

E) The correct answer is not given.

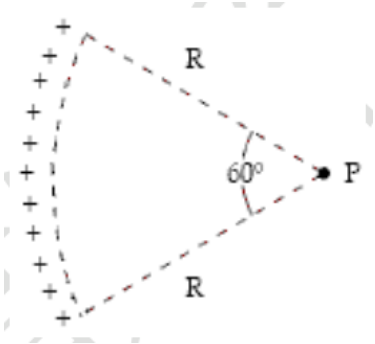
Answer: A

SOLUTION

$\vec{E} = -\nabla V = -6xz\,\hat{\imath} + 2z^3\,\hat{\jmath} - (3x^2 - 6yz^2)\,\hat{k}$. At (2,2,2): $-24\hat{\imath} + 16\hat{\jmath} - (12 - 48)\hat{k} = -24\hat{\imath} + 16\hat{\jmath} + 36\hat{k}$.

---

17 **Charge of uniform density 3.5 nC/m is distributed along the circular arc shown (radius R, angle 60°). Determine the potential at the center P.**



A) 61 V

B) 42 V

C) 52 V

D) 33 V ✓ (answer)

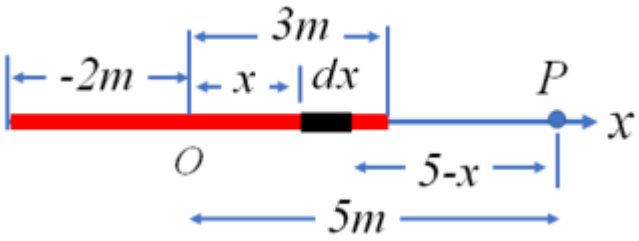
E) 22 V

Answer: D

SOLUTION

Every piece is the same distance R from P: $V = \frac{kq}{R} = \frac{k\lambda(R\theta)}{R} = k\lambda\theta = (9 \times 10^9)(3.5 \times 10^{-9})\frac{\pi}{3} = 33$ V. The radius cancels.

18 **10 nC is distributed uniformly along the x axis from $x = -2$ m to $x = +3$ m. Which integral gives the potential at $x = +5$ m?**



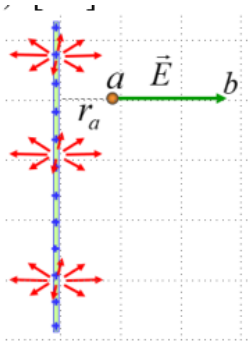
A) $\int_{-2}^{3} \frac{90\,dx}{x}$
B) $\int_{-2}^{3} \frac{90\,dx}{5-x}$
C) $\int_{-2}^{3} \frac{18\,dx}{x}$
D) $\int_{-2}^{3} \frac{18\,dx}{5-x}$ **✓ (answer)**
E) $\int_{-2}^{3} \frac{90\,dx}{5+x}$

Answer: D

SOLUTION

$\lambda = 10\ \text{nC}/5\ \text{m} = 2$ nC/m, so $k\lambda = 18$. A piece at x is $(5 - x)$ from the point: $dV = \frac{18\,dx}{5-x}$.

19 **Charge of uniform density 50 nC/m³ fills a long nonconducting cylindrical rod (radius 5.0 cm). Find $|V_A - V_B|$ between A (2.0 cm from the axis) and B (4.0 cm).**



A) 2.7 V

B) 2.0 V

C) 2.4 V

D) 1.7 V ✓ (answer)

E) 3.4 V

Answer: D

SOLUTION

Inside the rod, a coaxial Gaussian cylinder gives

$$E\,(2\pi r l) = \frac{\rho\,\pi r^2 l}{\varepsilon_0} \;\Rightarrow\; E = \frac{\rho r}{2\varepsilon_0}.$$

$$|V_A - V_B| = \int_{0.02}^{0.04} \frac{\rho r}{2\varepsilon_0}\,dr = \frac{\rho}{4\varepsilon_0}\left(0.04^2 - 0.02^2\right) = \frac{50\times10^{-9}}{4(8.85\times10^{-12})}(1.2\times10^{-3}) = 1.7\ \text{V}.$$

[NOTE] The official solution writes $E = \rho r/\varepsilon_0$ (missing the 2), but its final formula $\rho(b^2 - a^2)/4\varepsilon_0$ and its answer use the correct $\rho r/2\varepsilon_0$. **Why this method:** Gauss' law gives $E(r)$ from symmetry, then $\Delta V = -\int E\,dr$.

---

20 **A solid conducting sphere (radius 5.0 cm) has 0.25 nC on its surface. Point A is at the center and B is 15 cm from the center. What is $|V_A - V_B|$?**

A) 23 V

B) 30 V ✓ (answer)

C) 15 V

D) 45 V

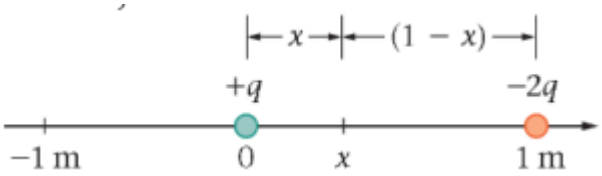
E) 60 V

Answer: B

SOLUTION

Inside a conductor V is constant, so $V_A = V_{\text{surface}} = \frac{kq}{0.05}$. Then $|V_A - V_B| = kq\left(\frac{1}{0.05} - \frac{1}{0.15}\right) = (9\times10^9)(0.25\times10^{-9})(13.3) = 30$ V.

Problem 1. A charge $q = 4.11 \times 10^{-9}$ C is at the origin and $-2q$ is on the x axis at $x = 1.0$ m. (a) Find the electric potential at $x = 0.50$ m. (b) The potential vanishes at some point between the charges. Find this x.



(a)

> **Answer:** $V = -74$ V

Superposition of scalars:

$$V = k\frac{q}{0.50} + k\frac{-2q}{0.50} = -k\frac{q}{0.50} = -\frac{(9.00 \times 10^{9})(4.11 \times 10^{-9})}{0.50} = -74 \text{ V}.$$

(The official solution writes –73.9 V.)

Why this method: Potentials add as plain numbers with signs, so no components are needed.

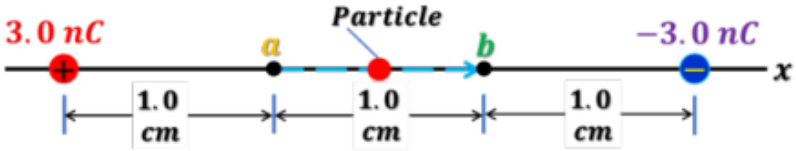
(b)

> **Answer:** $x = 0.33$ m

$$k\frac{q}{x} + k\frac{-2q}{1.00 - x} = 0 \;\Rightarrow\; 1.00 - x = 2x \;\Rightarrow\; x = \frac{1.00}{3} = 0.33 \text{ m}.$$

Why this method: Set the sum of potentials to zero and solve for x.

Problem 2. A dust particle ($m = 5.0 \times 10^{-9}$ kg, $q_0 = 2.0 \times 10^{-9}$ C) starts from rest and moves in a straight line from point a to point b (figure: $+3.0$ nC and -3.0 nC charges, points 1.0 cm apart as shown). What is its speed at b?



> **Answer:** $v_b = 46$ m/s

Only the conservative electric force acts, so mechanical energy is conserved:

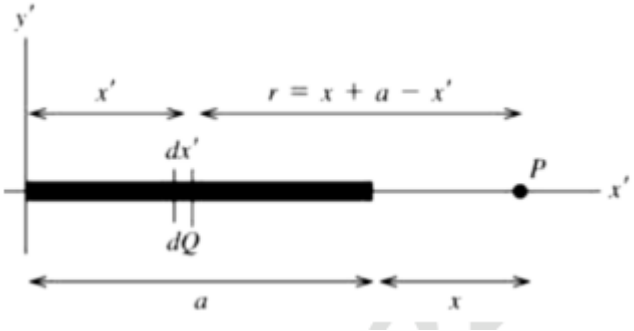
$$q_0 V_a = \tfrac{1}{2} m v_b^2 + q_0 V_b \;\Rightarrow\; v_b = \sqrt{\frac{2q_0(V_a - V_b)}{m}}.$$

$$V_a = k\left(\frac{3.0 \times 10^{-9}}{0.010} - \frac{3.0 \times 10^{-9}}{0.020}\right) = 1350 \text{ V}, \qquad V_b = -1350 \text{ V}.$$

$$v_b = \sqrt{\frac{2(2.0 \times 10^{-9})(2700)}{5.0 \times 10^{-9}}} = 46 \text{ m/s}.$$

Why this method: Energy conservation with $U = q_0 V$: no need to track the changing force along the way.

Problem 3. Charge Q is distributed uniformly along a thin rod of length a. Taking $V = 0$ at infinity, find the potential at a point a distance x to the right of the rod (on its axis).



> **Answer:** $V = \frac{Q}{4\pi\varepsilon_0 a}\ln\frac{x+a}{x}$

Slice the rod: $dQ = \frac{Q}{a}dx'$, at distance $r = x + a - x'$ from P.

$$V = \frac{Q}{4\pi\varepsilon_0 a}\int_0^a \frac{dx'}{x + a - x'} = \frac{Q}{4\pi\varepsilon_0 a}\ln\frac{x+a}{x}.$$

Check: as $x \to \infty$, $\ln\frac{x+a}{x} \approx \frac{a}{x}$, so $V \to \frac{Q}{4\pi\varepsilon_0 x}$, like a point charge.

Why this method: Continuous charge: integrate $dV = k\,dq/r$, a scalar integral.

Problem 4. In a region the potential is $V(x, y, z) = Axy - Bx^2 + Cy$, with A, B, C positive constants. (a) Find the x, y and z components of $\vec{E}$. (b) Where is $\vec{E} = 0$?

(a)

> **Answer:** $E_x = -Ay + 2Bx,\ E_y = -Ax - C,\ E_z = 0$

$$E_x = -\frac{\partial V}{\partial x} = -Ay + 2Bx, \qquad E_y = -\frac{\partial V}{\partial y} = -Ax - C, \qquad E_z = -\frac{\partial V}{\partial z} = 0.$$

Why this method: $\vec{E} = -\nabla V$, one partial derivative per component.

(b)

> **Answer:** $x = -C/A,\ y = -2BC/A^2$, any z

$E_y = 0 \Rightarrow x = -C/A$. Then $E_x = 0 \Rightarrow y = \frac{2Bx}{A} = -\frac{2BC}{A^2}$. $E_z = 0$ everywhere, so this holds on the whole line (any z).

Why this method: Set every component to zero and solve together.